

Lecture 1: Foundations

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Course of Probability and Statistics

Fall, 2026

Materials are taken from different sources with modifications. I do not claim any authorship of these materials.

Content

- Introduction to Probability
- Sample Spaces, outcomes and events
- Operations with events
- The probability of an event
- σ -Algebras
- Axioms of Probability

Introduction to Probability

Probability theory is the branch of mathematics that deals with the analysis of random phenomena.

The central objects of probability theory are

- random experiments
- outcomes
- events
- the likelihood of these events occurring

Examples of random experiments:

1. Tossing a coin

- outcomes: Heads (H) or Tails (T)
- events:
 - Simple Event (Getting Heads): $\{H\}$
 - Certain Event (Getting either Heads or Tails): $\{H, T\}$

2. Rolling a die:

- outcomes 1, 2, 3, 4, 5, or 6
- events:
 - Rolling an Even Number (Compound Event): $\{2, 4, 6\}$
 - Rolling a Number Greater than 4: $\{5, 6\}$

3. Measuring the height of a randomly selected student.

4. **Atmospheric Noise Sampling:** Record the exact millisecond value of a atmospheric static electrical pulse captured by a radio receiver.

Empirical (Relative Frequency) Probability

Definition

The probability of an event A is the proportion of times the event occurs in a large number of repeated trials of the experiment.

$$P(A) \approx \frac{\text{Number of times } A \text{ occurred}}{\text{Total number of trials}}$$

As the number of trials $n \rightarrow \infty$, the relative frequency stabilizes to the true probability.

Example

If a coin is tossed 1000 times and lands on Heads 510 times, the empirical probability of Heads is:

$$P(\text{Heads}) \approx \frac{510}{1000} = 0.51$$

The first column of this table indicates the experiment number; the subsequent columns contain data on the numbers m of heads obtained in n ($= 10^2, 10^4, 10^6$) tosses (trials) of a fair symmetric coin. There is an obvious stabilization of the relative frequency $p_n = m/n$ of heads as the number of trials n increases, as well as the convergence of p_n to the value $p = 1/2$.

$N \backslash n$	10^2	10^4	10^6
1	41	4985	499558
2	48	5004	499952
3	44	5085	500114
4	52	4946	500064
5	58	4978	500183
6	52	4985	499533
7	45	5012	500065
8	50	4931	500317
9	52	5016	500449
10	45	4973	500704
<i>Er</i>	10^{-1}	10^{-2}	10^{-3}

Sample Space

The discovered patterns, extended to trials with an arbitrary number of outcomes, make it possible to construct a simple mathematical model of a random experiment.

Sample Space

Definition (Sample Space)

The **sample space** S is the set of all possible outcomes of a random experiment.

- 1 Tossing a coin: $S = \{H, T\}$ (finite).
- 2 Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$ (finite).
- 3 Number of emails arriving at a server in an hour:
 $S = \{0, 1, 2, \dots\}$ (countably infinite).
- 4 Voltage across a resistor: $S = \mathbb{R}$ (uncountably infinite).
- 5 Time until a component fails: $S = [0, \infty)$ (uncountably infinite).
- 6 Temperature measurement: $S = \mathbb{R}$ (or $[0, \infty)$).
- 7 Position of a particle: $S = \mathbb{R}^3$ (uncountably infinite).

Example

Consider the experiment of tossing a die. If we are interested in the number that shows on the top face, the sample space is

$$S_1 = \{1, 2, 3, 4, 5, 6\}.$$

If we are interested only in whether the number is even or odd, the sample space is simply

$$S_2 = \{\text{even}, \text{odd}\}.$$

This example illustrates the fact that more than one sample space can be used to describe the outcomes of an experiment. In this case, S_1 provides more information than S_2 .

Definition (Event)

An **event** is any subset $A \subseteq S$. The event A occurs if the outcome ω satisfies $\omega \in A$. A collection of subsets of S we denote by $\mathcal{F}$.

Example

Given the sample space $S = \{t \mid t \geq 0\}$, where t is the life in years of a certain electronic component, then the event A that the component fails before the end of the fifth year is the subset $A = \{t \mid 0 \leq t < 5\}$.

It is conceivable that an event may be a subset that includes the entire sample space S or a subset of S called the **null** set and denoted by the symbol ϕ , which contains no elements at all. For instance, if we let A be the event of detecting a microscopic organism by the naked eye in a biological experiment, then $A = \phi$. Also, if

$$B = \{x \mid x \text{ is an even factor of } 7\},$$

then B must be the null set, since the only possible factors of 7 are the odd numbers 1 and 7.

Operations with events

We now consider certain operations with events that will result in the formation of new events. These new events will be subsets of the same sample space as the given events. Suppose that A and B are two events associated with an experiment. In other words, A and B are subsets of the same sample space S . For example, in the tossing of a die we might let A be the event that an even number occurs and B the event that a number greater than 3 shows. Then the subsets $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$ are subsets of the same sample space

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Note that both A and B will occur on a given toss if the outcome is an element of the subset $\{4, 6\}$, which is just the intersection of A and B .

Definition

The intersection of two events A and B , denoted by the symbol $A \cap B$, is the event containing all elements that are common to A and B .

Example

Let E be the event that a person selected at random in a classroom is majoring in engineering, and let F be the event that the person is female. Then $E \cap F$ is the event of all female engineering students in the classroom.

Definition

The **complement** of an event A with respect to S is the subset of all elements of S that are not in A . We denote the complement of A by the symbol A' or A^C .

Example

Consider the sample space

$$S = \{\text{book, cell phone, mp3, paper, stationery, laptop}\}.$$

Let $A = \{\text{book, stationery, laptop, paper}\}$. Then the complement of A is $A' = \{\text{cell phone, mp3}\}$.

Disjoint sets

Definition

Two events A and B are mutually exclusive, or disjoint, if $A \cap B = \phi$, that is, if A and B have no elements in common.

Example

A cable television company offers programs on eight different channels, three of which are affiliated with ABC, two with NBC, and one with CBS. The other two are an educational channel and the ESPN sports channel. Suppose that a person subscribing to this service turns on a television set without first selecting the channel. Let A be the event that the program belongs to the NBC network and B the event that it belongs to the CBS network. The events A and B have no programs in common. Therefore, the intersection $A \cap B$ contains no programs, and consequently the events A and B are mutually exclusive.

Often one is interested in the occurrence of at least one of two events associated with an experiment. Thus, in the die-tossing experiment, if

$$A = \{2, 4, 6\} \quad \text{and} \quad B = \{4, 5, 6\},$$

we might be interested in either A or B occurring or both A and B occurring. Such an event, called the **union** of A and B , will occur if the outcome is an element of the subset $\{2, 4, 5, 6\}$.

Union of the two events

Definition

The **union** of the two events A and B , denoted by the symbol $A \cup B$, is the event containing all the elements that belong to A or B or both.

Example

Let $A = \{a, b, c\}$ and $B = \{b, c, d, e\}$; then $A \cup B = \{a, b, c, d, e\}$.

Example

Let P be the event that an employee selected at random from an oil drilling company smokes cigarettes. Let Q be the event that the employee selected drinks alcoholic beverages. Then the event $P \cup Q$ is the set of all employees who either drink or smoke or do both. If $M = \{x \mid 3 < x < 9\}$ and $N = \{y \mid 5 < y < 12\}$, then $M \cup N = \{z \mid 3 < z < 12\}$.

- **Complement:** $A^c = \{\omega \in S : \omega \notin A\}$ (event that A does not occur).
- **Intersection:** $A \cap B = \{\omega \in S : \omega \in A \text{ and } \omega \in B\}$ (both events occur).
- **Union:** $A \cup B = \{\omega \in S : \omega \in A \text{ or } \omega \in B\}$ (at least one occurs).
- **Set difference:** $A \setminus B = \{\omega \in A : \omega \notin B\}$ (A occurs but not B).

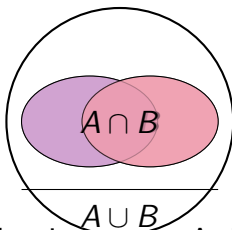


Figure 1: Venn diagram showing events A , B , intersection, and union.

The following equalities hold:

1. $A \cap \phi = \phi.$
2. $A \cup \phi = A.$
3. $A \cap A' = \phi.$
4. $A \cup A' = S.$
5. $S' = \phi.$
6. $\phi' = S.$
7. $(A')' = A.$
8. $(A \cap B)' = A' \cup B'.$
9. $(A \cup B)' = A' \cap B'.$

Probability

Let S be the sample space and $\mathcal{F}$ be the event space (σ -algebra). A probability measure P is a function $P : \mathcal{F} \rightarrow \mathbb{R}$ that satisfies the following three axioms:

- ① **Non-negativity:** For any event $A \in \mathcal{F}$,

$$P(A) \geq 0$$

- ② **Normalization:** For the certain event S ,

$$P(S) = 1$$

- ③ **Countable Additivity:** For any sequence of mutually exclusive (disjoint) events $A_1, A_2, A_3, \dots$ (where $A_i \cap A_j = \emptyset$ for $i \neq j$),

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Definition

The probability of an event A is the sum of the weights of all sample points in A . Therefore,

$$0 \leq P(A) \leq 1, \quad P(\phi) = 0, \quad \text{and} \quad P(S) = 1.$$

Furthermore, if $A_1, A_2, A_3, \dots$ is a sequence of mutually exclusive events, then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

Example

A coin is tossed twice. What is the probability that at least 1 head occurs? *Solution:* The sample space for this experiment is

$$S = \{HH, HT, TH, TT\}.$$

Solution: If the coin is balanced, each of these outcomes is equally likely to occur. Therefore, we assign a probability of ω to each sample point. Then $4\omega = 1$, or $\omega = \frac{1}{4}$. If A represents the event of at least 1 head occurring, then

$$A = \{HH, HT, TH\} \text{ and } P(A) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}.$$

Example

A die is loaded in such a way that an even number is twice as likely to occur as an odd number. If E is the event that a number less than 4 occurs on a single toss of the die, find $P(E)$.

Solution: The sample space is $S = \{1, 2, 3, 4, 5, 6\}$. We assign a probability of w to each odd number and a probability of $2w$ to each even number. Since the sum of the probabilities must be 1, we have $9w = 1$ or $w = 1/9$. Hence, probabilities of $1/9$ and $2/9$ are assigned to each odd and even number, respectively. Therefore,

$$E = \{1, 2, 3\} \text{ and } P(E) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} = \frac{4}{9}.$$

Example

Let A be the event that an even number turns up and let B be the event that a number divisible by 3 occurs. Find $P(A \cup B)$ and $P(A \cap B)$.

Solution: For the events $A = \{2, 4, 6\}$ and $B = \{3, 6\}$, we have

$$A \cup B = \{2, 3, 4, 6\} \text{ and } A \cap B = \{6\}.$$

By assigning a probability of $1/9$ to each odd number and $2/9$ to each even number, we have

$$P(A \cup B) = \frac{2}{9} + \frac{1}{9} + \frac{2}{9} + \frac{2}{9} = \frac{7}{9} \quad \text{and} \quad P(A \cap B) = \frac{2}{9}.$$

If the sample space for an experiment contains N elements, all of which are equally likely to occur, we assign a probability equal to $1/N$ to each of the N points.

If an experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then the probability of event A is

$$P(A) = \frac{n}{N}.$$

Example

A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is (a) an industrial engineering major and (b) a civil engineering or an electrical engineering major.

Solution: Denote by I , M , E , and C the students majoring in industrial, mechanical, electrical, and civil engineering, respectively. The total number of students in the class is 53, all of whom are equally likely to be selected.

(a) Since 25 of the 53 students are majoring in industrial engineering, the probability of event I , selecting an industrial engineering major at random, is

$$P(I) = \frac{25}{53}.$$

(b) Since 18 of the 53 students are civil or electrical engineering majors, it follows that $P(C \cup E) = \frac{18}{53}$.

Theorem

If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof.

The $P(A \cup B)$ is the sum of the probabilities of the sample points in $A \cup B$. Now $P(A) + P(B)$ is the sum of all the probabilities in A plus the sum of all the probabilities in B . Therefore, we have added the probabilities in $(A \cap B)$ twice. Since these probabilities add up to $P(A \cap B)$, we must subtract this probability once to obtain the sum of the probabilities in $A \cup B$. □

Corollary

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

Proof.

The corollary is an immediate result of the theorem, since if A and B are mutually exclusive, $A \cap B = \emptyset$ and then $P(A \cap B) = P(\emptyset) = 0$. In general, we can write the Corollary. □

Example

John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company A is 0.8, and his probability of getting an offer from company B is 0.6. If he believes that the probability that he will get offers from both companies is 0.5, what is the probability that he will get at least one offer from these two companies?

Solution

Using the additive rule, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.6 - 0.5 = 0.9.$$

Example

What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?

Solution

Let A be the event that 7 occurs and B the event that 11 comes up. Now, a total of 7 occurs for 6 of the 36 sample points, and a total of 11 occurs for only 2 of the sample points. Since all sample points are equally likely, we have $P(A) = 1/6$ and $P(B) = 1/18$. The events A and B are mutually exclusive, since a total of 7 and 11 cannot both occur on the same toss. Therefore,

$$P(A \cup B) = P(A) + P(B) = \frac{1}{6} + \frac{1}{18} = \frac{2}{9}.$$

Theorem (Inclusion-Exclusion for Three Events)

For any events A, B, C ,

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

Property 1: Probability of the Empty Set

$$\mathbb{P}(\emptyset) = 0.$$

Property 2: Finite Additivity For disjoint $A_1, \dots, A_n$:

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i).$$

Property 3: Complement

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

Property 4: Monotonicity: If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Property 5: Union Bound

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

Theorem

If A and A' are complementary events, then

$$P(A) + P(A') = 1.$$

Proof.

Since $A \cup A' = S$ and the sets A and A' are disjoint,

$$1 = P(S) = P(A \cup A') = P(A) + P(A').$$



Example

If the probabilities that an automobile mechanic will service 3, 4, 5, 6, 7, or 8 or more cars on any given workday are, respectively, 0.12, 0.19, 0.28, 0.24, 0.10, and 0.07, what is the probability that he will service at least 5 cars on his next day at work?

Solution: Let E be the event that at least 5 cars are serviced. Now,

$$P(E) = 1 - P(E'),$$

where E' is the event that fewer than 5 cars are serviced. Since

$$P(E') = 0.12 + 0.19 = 0.31,$$

it follows from the theorem that

$$P(E) = 1 - 0.31 = 0.69.$$

Definition

If a random experiment can result in n equally likely outcomes, and if event A can occur in exactly m of these outcomes, then the probability of event A is:

$$P(A) = \frac{m}{n} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Example

What is the probability of rolling a 4 on a fair six-sided die?

$$P(4) = \frac{1}{6}$$

Limitation: Only works when all outcomes are equally likely.

Geometric probability

Let S be a sample space that can be represented as a measurable subset of $\mathbb{R}^n$ (length, area, or volume).

Definition (Geometric Probability)

Let $A \subseteq S$ be an event of interest. If all outcomes in S are equally likely, then the probability of event A is given by the ratio

$$P(A) = \frac{\text{measure}(A)}{\text{measure}(S)}$$

where the measure is:

- **length** if $S \subset \mathbb{R}$,
- **area** if $S \subset \mathbb{R}^2$,
- **volume** if $S \subset \mathbb{R}^3$.

Geometric probability

Example

Two friends, Alice and Bob, agree to meet at a certain location between 12:00 and 13:00. Each arrives independently at a uniformly random time during the hour. They agree to wait for exactly 15 minutes. What is the probability that they meet?

Geometric probability

Let x be Alice's arrival time in minutes after 12:00 ($0 \leq x \leq 60$).

Let y be Bob's arrival time in minutes after 12:00 ($0 \leq y \leq 60$).

Since they arrive independently and uniformly, the sample space is a square of size 60×60 in the Cartesian plane. The total area of this sample space is:

$$\text{Total Area} = 60 \times 60 = 3600$$

Alice and Bob will meet if the absolute difference between their arrival times is at most 15 minutes:

$$|x - y| \leq 15$$

This inequality can be rewritten as a system of two linear inequalities:

$$x - 15 \leq y \leq x + 15$$

Geometric probability

The region where they meet is shaded in the figure below. It lies between the boundary lines $y = x + 15$ and $y = x - 15$.

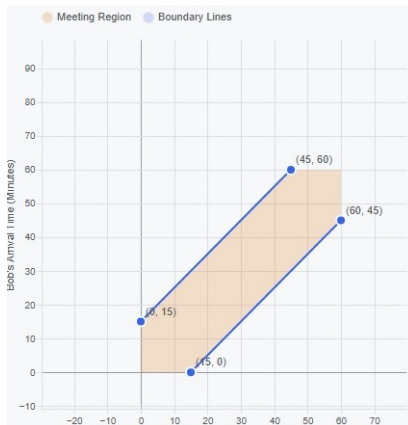


Figure: Geometric Probability Space for the Meeting Problem.

Geometric probability

Each of the two unshaded triangles has a base and height of $60 - 15 = 45$ minutes. Combined, their total area is:

$$\text{Non-Meeting Area} = 2 \times \left(\frac{1}{2} \times 45 \times 45 \right) = 45 \times 45 = 2025$$

Subtracting the non-meeting area from the total area yields the favorable meeting area:

$$\text{Meeting Area} = 3600 - 2025 = 1575$$

The probability P that Alice and Bob meet is the ratio of the meeting area to the total sample space area:

$$P(\text{Meeting}) = \frac{\text{Meeting Area}}{\text{Total Area}} = \frac{1575}{3600} = \frac{7}{16} = 0.4375$$

Thus, the probability that they successfully meet is $\frac{7}{16}$ **or** 43.75%.

σ -Algebras: Why We Need Them

In many cases, not every subset of S can be assigned a probability.

Definition (σ -Algebra)

A collection $\mathcal{F}$ of subsets of S is a σ -**algebra** if:

- 1 $S \in \mathcal{F}$.
- 2 If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (closed under complements).
- 3 If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closed under countable unions).

The pair $(S, \mathcal{F})$ is called a **measurable space**.

Examples of σ -Algebras

Let X be a non-empty set.

1. Trivial (Minimal) σ -Algebra

$$\mathcal{F}_0 = \{\emptyset, X\}$$

2. Power Set (Maximal) σ -Algebra

$$\mathcal{F}_{\max} = 2^X = \mathcal{P}(X)$$

3. For any non-empty proper subset $A \subset X$:

$$\mathcal{F}_A = \{\emptyset, A, A^c, X\}$$

4. Borel σ -Algebra on $\mathbb{R}$

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b) : a < b, a, b \in \mathbb{R}\})$$

5. Countable-Cocountable σ -Algebra. For an uncountable set X :

$$\mathcal{F} = \{A \subseteq X : A \text{ is countable or } A^c \text{ is countable}\}$$

Examples of σ -Algebras

Let $X = \{a, b, c, d, e\}$. 1. Trivial σ -Algebra

$$\mathcal{F}_1 = \{\emptyset, X\}$$

2. Power Set (All subsets)

$$\mathcal{F}_2 = \mathcal{P}(X) = 2^X$$

This has $2^5 = 32$ elements.

3. Generated by a Single Element For $A = \{a, b\}$:

$$\mathcal{F}_3 = \{\emptyset, \{a, b\}, \{c, d, e\}, X\}$$

4. Generated by Two Elements For $A = \{a, b\}$ and $B = \{b, c\}$:

$$\mathcal{F}_4 = \{\emptyset, \{a, b\}, \{b, c\}, \{a, b, c\}, \{d, e\}, \{a, d, e\}, \{c, d, e\}, X\}$$

(where we take all unions and complements of A and B).

5. Generated by a Partition into Singletons

$$\mathcal{F}_5 = \mathcal{P}(X)$$

Definition (Probability Measure)

A function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a **probability measure** if it satisfies:

- 1 **Non-negativity:** $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$.
- 2 **Normalisation:** $\mathbb{P}(S) = 1$.
- 3 **Countable Additivity:** For any countable collection of mutually disjoint events $A_1, A_2, \dots$ ($A_i \cap A_j = \emptyset$ for $i \neq j$),

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(S, \mathcal{F}, \mathbb{P})$ is a **probability space**.

Axiom 1: Probabilities are never negative.

Axiom 2: Something must happen with probability 1.

Axiom 3: The probability of a union of disjoint events is the sum of their probabilities. This generalises the idea that if events cannot happen together, their probabilities add.

Axioms of Probability (Kolmogorov)

These are the fundamental rules that any probability function $P(\cdot)$ must satisfy.

Axiom 1: Non-Negativity

For any event A ,

$$P(A) \geq 0$$

Axiom 2: Normalization

The probability of the sample space is 1,

$$P(S) = 1$$

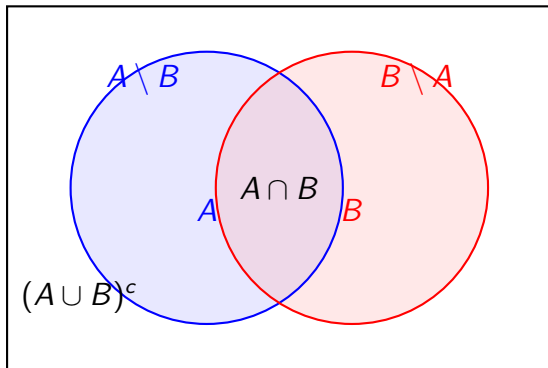
Axiom 3: Additivity

For any mutually exclusive (disjoint) events $A_1, A_2, \dots$,

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

Visualizing Axioms (Venn Diagrams)

S



Note: Additivity (Axiom 3) requires $A \cap B = \emptyset$ (no overlap).

Summary

- A **Random Experiment** has outcomes that are not predetermined.
- The **Sample Space** is the set of all possible outcomes.
- An **Event** is a subset of the sample space.
- The three **Axioms** (Non-negativity, Normalization, Additivity) form the foundation of probability theory.
- Probability is a powerful tool for quantifying uncertainty.

Thank You!